
AI-Driven Inverse Design of Thermoelectric Materials Focusing on Electrical Conductivity Using Public Materials Databases

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Abstract

Thermoelectric materials have attracted significant attention due to their ability to directly convert temperature gradients into electrical energy, enabling waste heat recovery and improved energy efficiency. However, the optimization of thermoelectric performance is challenging because the key properties—Seebeck coefficient, electrical conductivity, and thermal conductivity—exhibit strong trade-off relationships. In this study, electrical conductivity (σ) was selected as a single target property to reduce complexity, and an AI-based inverse design approach was applied using composition-based descriptors. Publicly available thermoelectric datasets were integrated, and records containing compositional information, temperature, and electrical conductivity were selected. Compositional features were transformed into elemental fraction vectors and statistical descriptors, which were used to train an AI model for predicting electrical conductivity. Based on the trained model, inverse design exploration was performed in the compositional space to identify candidate compositions with high predicted electrical conductivity at the commercially relevant and data-rich temperature of 700 K. Prediction uncertainty, number of constituent elements, and chemical screening criteria were additionally considered to organize and rank the candidate set. The resulting top-ranked candidates predominantly consisted of relatively simple compositions with two to three elements and exhibited continuity within specific compositional families. These results demonstrate that inverse design based on a single target property can effectively narrow the compositional search space and identify promising candidates. This study provides a practical foundation for stepwise AI-driven thermoelectric material design, with potential extensions to multi-property optimization and structural information integration.

1 Introduction

Thermoelectric materials are functional materials capable of directly converting temperature gradients into electrical energy and have therefore attracted significant interest for waste heat recovery and energy efficiency improvement. Thermoelectric performance is commonly evaluated using the dimensionless figure of merit, $ZT = S^2\sigma T/\kappa$, where the Seebeck coefficient (S), electrical conductivity (σ), and thermal conductivity (κ) exhibit strong trade-off relationships. As a result, improving a single property often leads to the degradation of others, making the optimization of thermoelectric performance a long-standing and challenging problem. Conventional exploration of thermoelectric materials has relied

primarily on experimental composition tuning or computational screening based on density functional theory (DFT). However, thermoelectric materials possess high-dimensional design spaces involving composition, crystal structure, defect states, and doping concentrations, leading to an exponential increase in the number of possible candidates. In such high-dimensional search spaces, traditional approaches suffer from clear limitations in terms of time and computational cost. Recently, data-driven approaches leveraging artificial intelligence (AI) have gained considerable attention in materials science as a means to overcome these limitations. In particular, inverse design strategies, in which a forward model is first constructed to predict material properties from compositional information and then used to identify compositions that satisfy target properties, are regarded as a paradigm shift in materials design. These approaches enable systematic exploration of compositional regions that are difficult to access through human intuition alone. In this study, we construct a composition-based AI prediction model using publicly available thermoelectric materials databases and perform inverse design exploration with electrical conductivity as the target property. Although the databases contain various thermoelectric properties, including the Seebeck coefficient, thermal conductivity, and ZT, the present work restricts the target property to electrical conductivity to ensure a stable compositional input representation and a consistent inverse design pipeline. Through this approach, we aim to demonstrate that AI-based inverse design can be extended to the practical stage of candidate composition identification. Furthermore, this study was conducted collaboratively by a first-year high school student and a supervising teacher, demonstrating that AI-based approaches to advanced materials design problems can yield meaningful research outcomes even within an educational context. By employing AI throughout the entire research workflow, from data curation and feature extraction to model training and candidate proposal, we present AI not merely as a computational tool but as a collaborative co-scientist supporting the research process. This work thereby examines the openness, reproducibility, and learnability of AI-assisted research in materials science.

2 Experimental

2.1 Data collection and preparation

In this study, publicly accessible experimental and computational thermoelectric materials databases were utilized to predict the electrical conductivity of thermoelectric materials. The collected data consisted primarily of records that included compositional formulas and measured or calculated temperatures, along with electrical conductivity values. Although the databases also contained additional thermoelectric properties such as the Seebeck coefficient, thermal conductivity, and ZT, the present study focused on electrical conductivity as the target property for analysis. Data collected from different sources exhibited variations in unit systems, temperature notation, and compositional formula formats; therefore, normalization and data cleaning procedures were applied prior to integration. Electrical conductivity values were standardized to units of S/m, and temperatures were converted to absolute temperature (K). Compositional formulas were unified using chemical formulas expressed by elemental symbols and stoichiometric coefficients, while records with nonstandard or incomplete compositional expressions were excluded during the cleaning process. Duplicate records were identified based on identical compositional formulas and temperature conditions, and entries with inconsistent property values among duplicates were excluded from further analysis. In addition, any records lacking compositional formula, temperature, or electrical conductivity information were removed. Through these procedures, a dataset suitable for composition-based electrical conductivity prediction was constructed.

2.2 Input feature design

The input variables for the AI model were constructed using composition-based descriptors to numerically represent compositional information. Each compositional formula was decomposed into its constituent elements and converted into an elemental fraction vector relative to the overall composition. Based on these vectors, average properties of the constituent elements, including mean atomic number, mean atomic mass, and mean electronegativity, were calculated. To capture the inhomogeneity of elemental distributions within each composition, statistical descriptors such as variance, maximum–minimum range, and standard deviation of atomic number, atomic mass, and electronegativity were additionally included. These features were used to quantitatively represent compositional complexity

and heterogeneity. In order to ensure a consistent input dimensionality across all samples, crystallographic structural information was excluded from the input variables. Consequently, all records were represented solely by composition-based descriptors, and no property values, including electrical conductivity, were included in the input features. The resulting input matrix was used in a consistent format throughout the model training and prediction stages.

2.3 Construction of the electrical conductivity prediction model

An electrical conductivity prediction model was constructed using the composition-based input features and corresponding electrical conductivity values. The model was defined to take compositional information and temperature conditions as inputs and output electrical conductivity values, with electrical conductivity specified as the single target property. Considering the distribution characteristics of electrical conductivity, model training and analysis were performed using logarithmic-scale electrical conductivity values. The dataset was divided into training and validation subsets, and model performance was evaluated on both the training data and independent validation data. Prediction results were compared on the basis of log-scale electrical conductivity, and repeated training procedures were conducted to confirm the reproducibility of model predictions. The trained model was subsequently used as an evaluation function for electrical conductivity in the inverse design stage.

2.4 Electrical conductivity-based inverse design strategy

Based on the trained electrical conductivity prediction model, an inverse design strategy was applied to explore compositions with high electrical conductivity. During the inverse design process, new compositional candidates were generated within the compositional space, and electrical conductivity values were predicted for each candidate. The generated candidates were then ranked according to their predicted electrical conductivity. Candidate evaluation considered not only predicted electrical conductivity values but also auxiliary indicators such as prediction uncertainty, number of constituent elements, and the presence of rare elements. These indicators were recorded for each candidate, and a set of top-ranked candidate compositions was constructed based on electrical conductivity predictions. Candidate compositions were managed at the level of compositional formulas, and when multiple temperature conditions were predicted for the same composition, the results were treated separately according to temperature.

2.5 Quantification of prediction uncertainty

To assess the reliability of electrical conductivity predictions, prediction uncertainty was quantified. For identical compositional and temperature conditions, repeated predictions were performed, and the variance of the predicted results was calculated and defined as an uncertainty metric. The resulting uncertainty values were recorded together with the predicted electrical conductivity for each record. Prediction uncertainty was used as an auxiliary indicator in the ranking and classification of candidate compositions, enabling quantitative comparison among candidates with similar electrical conductivity values. The uncertainty metric was further used to analyze the characteristics of the final candidate set and is presented together with the distribution of candidate compositions in the Results section.

3 Results

3.1 Integrated dataset and identification of valid records

The integrated data table consisted of a total of 46,519 rows and 42 columns. Among these, 5,205 records were identified in which the compositional formula, temperature, electrical conductivity, Seebeck coefficient, thermal conductivity, and ZT were all provided. Within the valid records (5,205 entries), the compositional formula, temperature (K), and electrical conductivity (S/m) were available for all records, and no missing values were observed for electrical conductivity. In addition, the Seebeck coefficient, thermal conductivity, and ZT were present in a large fraction of

the same records. Accordingly, within the dataset, the subsequent data processing steps were governed not by the presence or absence of individual property values, but by whether the compositional formula could be expressed in a numerical form suitable for analysis. With respect to metadata, columns such as reference identifiers, data type, and sample form were included at the record level, enabling classification according to data origin (e.g., experimental or computational) and sample form (e.g., bulk or film). Although derived property columns such as power factor were also included in the dataset, the Results section focuses on variables and processing steps directly used for electrical conductivity prediction and inverse design.

3.2 Compositional parsing and determination of the analysis dataset

Compositional formulas in the valid records (5,205 entries) were processed by decomposing each formula into elemental components and converting them into elemental fraction vectors. As a result, compositional parsing was successfully completed for 4,521 records. Records with successful parsing were included in subsequent input construction and model training, while records for which parsing was not possible were excluded from further analysis. Consequently, 4,521 records, corresponding to 86.9% of the valid dataset, were confirmed as convertible into model-ready inputs. The success or failure of compositional parsing was observed to be independent of the presence of property values. That is, even when electrical conductivity and other thermoelectric properties were provided, records with compositional expressions that could not be uniquely decomposed into elemental fraction vectors were excluded at the parsing stage. The finalized set of 4,521 records was used as the reference dataset for subsequent analyses, including input matrix construction, prediction result generation, and inverse design candidate identification. Table 1 summarizes the number of valid records and successfully parsed records.

Table 1: Summary of Valid and Parsable Records in the Integrated Thermoelectric Dataset

Category	Number of records
Valid records with complete property and temperature information	5,205
Records with successfully parsed compositions (analysis dataset)	4,521
Records excluded due to unparsable compositions	684

3.3 Construction of the composition-based descriptor matrix

For the 4,521 records with successfully parsed compositions, composition-based input variables were generated, and each record was converted into a fixed-dimensional numerical vector. The resulting input matrix consisted of 4,521 rows and 117 columns. The input variables were derived solely from compositional information and included elemental fraction vectors as well as statistical summary descriptors calculated from the composition, such as atomic-number-related statistics. No property values were included in the input matrix, and electrical conductivity was retained exclusively as the target variable. Consequently, the input–output structure was configured to avoid data leakage arising from variable overlap. The descriptor matrix was standardized to have an identical column structure across all records, ensuring a consistent input format for subsequent training and prediction stages. The structural characteristics of the input matrix are summarized below.

Table 2: Summary of the Compositional Descriptor Matrix Used for Electrical Conductivity Prediction

Item	Value
Number of samples	4,521
Number of input features	117

Input feature type	Composition-based numerical descriptors
Output variable	Electrical conductivity (σ)

3.4 Construction of the electrical conductivity training dataset

To predict electrical conductivity, a training dataset was constructed by combining the composition-based descriptor matrix (4,521 samples) with the corresponding electrical conductivity values. Each sample in the training dataset was defined by an input vector corresponding to a specific composition and temperature condition, together with an electrical conductivity output value. The size of the training dataset was therefore fixed at 4,521 samples, identical to that of the input matrix. Compositional formulas and temperature information were retained within the training dataset, enabling identification of the predicted conditions in subsequent analyses. Metadata such as data type (experimental or computational) and sample form, which were present in the original dataset, were also preserved in a record-traceable manner in the constructed training table.

3.5 Electrical conductivity prediction results and candidate identification

Electrical conductivity prediction was performed using the constructed training dataset, and the resulting prediction table consisted of 1,744 rows and 6 columns. Predictions were generated for 872 distinct compositional formulas, with two temperature conditions (300 K and 700 K) evaluated for each composition, yielding 872 predictions per temperature and a total of 1,744 prediction records. The prediction results included logarithmic electrical conductivity values, their corresponding standard deviations, and electrical conductivity values expressed on a linear scale. Each prediction record also contained ranking information, reflecting the ordering of candidates based on predicted values. From the prediction results, candidate compositions at the 700 K condition were selected, yielding a candidate set consisting of 88 compositions. Each record in the candidate set included the predicted electrical conductivity at 700 K, prediction uncertainty metrics, a rare-element penalty (rare_penalty), and a composite score (score). Ranking and prioritization information for the candidate compositions were provided through the rank field. A summary of the prediction results and candidate set is given below.

Table 3: Summary of Electrical Conductivity Prediction Results and Candidate Selection

Item	
Item	Value
Number of predicted compositions	872
Prediction temperatures	300 K, 700 K
Number of prediction records	1,744
Number of candidate compositions at 700 K	88

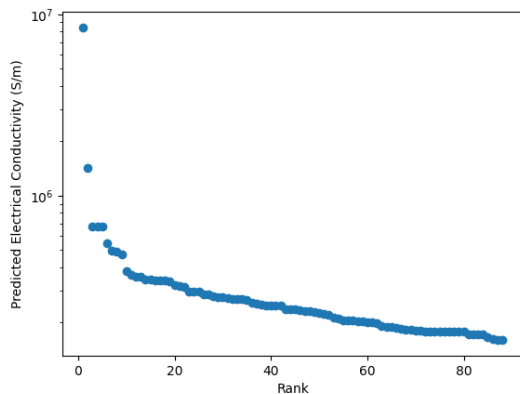


Figure 1: Rank versus predicted electrical conductivity of inverse-designed candidates

Figure 1 illustrates the relationship between inverse design rank and predicted electrical conductivity (σ) for the entire set of candidate compositions generated through the AI-based inverse design process. This plot includes the full candidate pool prior to any additional filtering, with the ranking extending beyond rank 80. Several top-ranked candidates exhibit relatively high electrical conductivity values, while a gradual decrease in conductivity is observed as the rank increases. This distribution reflects the stepwise ordering of candidate compositions based on predicted electrical conductivity during the inverse design process.

3.6 Extension of auxiliary indicators and distribution characteristics of candidate compositions

Auxiliary indicators were expanded for the 88 candidate compositions, and the candidate set was organized into a table consisting of 88 rows and 14 columns. In addition to the primary predicted values of electrical conductivity (σ) and prediction uncertainty, the table included indicators such as training data inclusion status, uncertainty category, number of constituent elements, and flags related to chemical exclusion criteria. The distribution of the number of constituent elements among the candidate compositions was concentrated in the range of two to five elements, with 16 binary, 31 ternary, 28 quaternary, and 13 quinary compositions identified. Prediction uncertainty categories were distributed as 30 high, 28 medium, and 30 low uncertainty candidates. With respect to training data inclusion, 77 candidates were identified as present in the training dataset, while 11 candidates were not. The distribution of chemical exclusion flags was as follows: `toxic_flag = True` for 7 candidates, `contains_halogen = True` for 6 candidates, and `contains_oxygen = True` for 1 candidate. These flags were provided as record-level indicators to support further classification and exclusion criteria for candidate compositions.

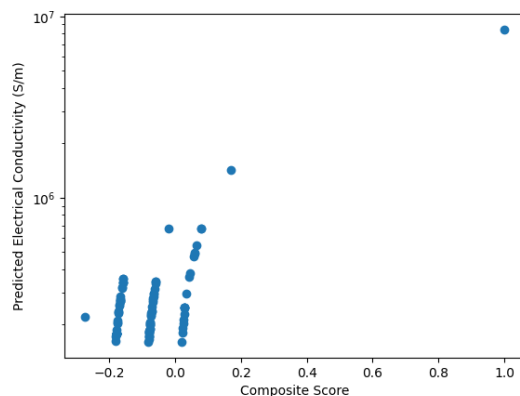


Figure 2: Distribution of inverse-designed candidates in the composite score- σ space

Figure 2 presents the inverse-designed candidate compositions in terms of their composite score and predicted electrical conductivity (σ). Rather than forming a continuous distribution, the candidates exhibit clustered distributions along the score axis. As the composite score incorporates

not only electrical conductivity but also prediction uncertainty and composition-related constraint terms, this distribution visually demonstrates that the inverse design process was guided by multiple criteria rather than a single-property objective.

3.7 Table of top-ranked electrical conductivity candidate compositions

Based on the electrical conductivity prediction results, candidate compositions with high electrical conductivity at 700 K were selected. The candidates were sorted in descending order according to predicted electrical conductivity, and identical ranks were assigned to compositions with the same predicted conductivity values. Table 4 summarizes the compositional formulas, predicted electrical conductivity (σ), logarithmic-scale conductivity values, prediction uncertainties, and numbers of constituent elements for the top-ranked candidates. The highest-ranked candidates predominantly consisted of two- to three-element compositions and included metal–chalcogenide systems (Co–S, Cu–S, Cu–Se) as well as metal–silicide systems (Co–Si). Compositions at the top of the ranking exhibited electrical conductivity values on the order of 10^6 S/m, with a gradual decrease in predicted conductivity observed as the rank increased. Several compositions with identical predicted conductivity values involved compositional modifications through minor elemental substitution; in these cases, identical conductivity values were obtained. Prediction uncertainty varied among candidate compositions, with some candidates showing relatively low uncertainty and others exhibiting higher uncertainty. These metrics were provided alongside electrical conductivity values to enable quantitative comparison of candidate characteristics.

Table 4: Top-Ranked Inverse-Designed Candidate Compositions with High Electrical Conductivity at 700 K

Rank	Composition	Predicted σ (S/m)	$\log_{10}(\sigma)$	Uncertainty (std)	Number of elements
1	Co ₉ S ₈	8.44×10^6	6.93	0.224	2
2	CoS ₂	1.42×10^6	6.15	0.125	2
3	CoSi	6.75×10^5	5.83	0.345	2
3	CoSi _{0.995} Bo _{0.005}	6.75×10^5	5.83	0.345	3
4	CoSi _{0.98} Ge _{0.02}	6.73×10^5	5.83	0.508	3
5	Cu _{1.7} Se	5.46×10^5	5.74	0.094	2
6	Cu _{1.5} S	4.96×10^5	5.70	0.023	2
7	Cu _{1.8} Se	4.90×10^5	5.69	0.103	2
8	Cu _{1.55} S	4.73×10^5	5.67	0.024	2
9	Cu ₅ Sn ₂ Se ₇	3.83×10^5	5.58	0.097	3

4 Discussion

4.1 Significance of single-property inverse design based on electrical conductivity (σ)

Electrical conductivity directly reflects charge transport behavior and plays a central role in thermoelectric material design. In particular, when only composition-based inputs are used, electrical conductivity represents a property that can be predicted with relatively high stability, allowing clear trends associated with compositional variations to be captured. The top-ranked candidate compositions identified in this study exhibited a consistent ordering when ranked by predicted electrical conductivity, indicating that the composition-based AI model meaningfully

learned the relationship between composition and electrical conductivity. Accordingly, inverse design based on a single target property of electrical conductivity should be interpreted not as a final stage of thermoelectric material design, but as an initial step for effectively narrowing the compositional search space and identifying promising candidates. From this perspective, the present study demonstrates that electrical-conductivity-centered inverse design can be practically extended to the stage of candidate composition identification.

4.2 Common characteristics of AI-proposed candidate compositions

The top-ranked candidate compositions derived from inverse design exhibited several common characteristics. First, most high-ranking candidates consisted of relatively simple compositions composed of two to three elements, indicating that high electrical conductivity was predicted predominantly in regions of moderate compositional complexity. This trend suggests that the composition-based model provided more stable predictions in simpler compositional domains. Second, the top candidates showed a tendency to concentrate within specific compositional families, including metal–chalcogenide systems and metal–silicide systems. This observation reflects the relatively sufficient representation of these families in the training dataset and highlights the influence of data distribution on composition-based prediction models. Candidates involving minor doping or compositional substitution were also included among the top-ranked compositions; however, these candidates remained within the compositional continuity of the corresponding parent systems. Third, the distribution of prediction uncertainty varied among the top candidates. Even compositions with high predicted electrical conductivity exhibited differing uncertainty levels, demonstrating that electrical conductivity alone is insufficient to fully assess candidate reliability. This finding supports the necessity of incorporating prediction uncertainty metrics alongside predicted property values when evaluating inverse-designed candidates.

4.3 Limitations and future perspectives

This study was conducted under constraints related to limited time and data structure, which resulted in several inherent limitations. First, although the dataset contained multiple thermoelectric properties, the analysis focused on electrical conductivity as a single target property to ensure consistency in composition-based inputs and stability of the inverse design pipeline. Consequently, direct optimization or discussion of ZT or simultaneous multi-property optimization lies beyond the scope of this work. Second, by relying exclusively on composition-based inputs, factors that may significantly influence charge transport, such as crystal structure, defect states, and microstructural features, were not considered. Therefore, the candidate compositions proposed in this study should be interpreted as reflecting compositional-level potential rather than guaranteed realizations of target properties, and further structural considerations are required for practical implementation. Third, candidate compositions generated through inverse design are inevitably influenced by the distribution of the training data, resulting in a tendency for candidates to concentrate in compositional regions with relatively abundant data. This limitation may be alleviated in future studies through dataset expansion and more balanced sampling of the compositional space. Overall, this study demonstrates that AI-based inverse design centered on a single property of electrical conductivity can be practically extended to candidate composition identification, while simultaneously revealing the need for further research incorporating multi-property optimization and structural information. These findings suggest that AI-driven materials design can be progressively expanded through a stepwise and modular approach.

5 Conclusion

In this study, an AI prediction model based on compositional information was constructed using publicly available thermoelectric materials databases, and inverse design exploration was performed with electrical conductivity (σ) as the single target property. By integrating data curation, composition-based feature generation, electrical conductivity prediction, and inverse design candidate identification into a unified pipeline, this work confirms that AI-based approaches can be extended to the practical stage of proposing candidate compositions. Focusing on electrical conductivity as a single target property enabled effective reduction of the compositional search

space while avoiding the complexity of multi-property optimization. The top-ranked candidate compositions exhibited relatively simple compositional structures and continuity within specific compositional families, indicating that the composition-based AI model meaningfully captured the relationship between composition and electrical conductivity. Furthermore, incorporating auxiliary indicators such as prediction uncertainty, number of constituent elements, and chemical exclusion criteria allowed the construction of a comparable and interpretable candidate set rather than relying solely on predicted values. This multi-indicator-based organization enhances the interpretability of inverse design results and provides a foundation for subsequent validation and extended studies. This study shows that inverse design centered on electrical conductivity serves as an effective strategy for candidate screening prior to full multi-property optimization in thermoelectric material design, while also clarifying the limitations associated with composition-only inputs and data distribution dependence. Collectively, the results provide experimental evidence that AI-driven materials design can yield systematic candidate identification even under constrained conditions and establish a basis for stepwise expansion toward the integration of structural information and multi-property optimization.

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